EXPERIMENT-2

AIM: CONERT the ER diagram into a relational schema. Write Complete MySQL Create table statements with primary keys, foreign keys, Not null, unique, on delete cascade and set null constraints. Insert sample data and demonstrate referential integrity violations.

SOURCE CODE:

```
Create database ecommerce_db;
Use ecommerce-db;
--1 customer (strong entity)
Create table Customer (
    Customer-id  int auto_increment Primary key,
    First_name  Varchar(50) not null,
    Last_name Varchar(50) not null,
    email Varchar(50) not null unique
)

-- Multi Valued attribute : Customer phone numbers

Create Table Customer-Phone (
 customer-id  int not null,
 phone  varchar(50) not null,
 Primary key (customer-id, phone),
 Foreign key (customer-id) references Customer(customer-id)
);
```

--2 Address (weak entity depends on customer)

```
Create table Address(
    customer_id int not null,
    address_id int not null,
    street varchar(100) not null,
    city varchar(50) not null,
    State varchar(50) not null,
    pincode varchar(7) not null,
    Primary key (customer_id , address_id),
    Foreign key (customer_id) references customer(customer_id)
);
```

--3 Category

```
Create table Category (
    category_id int auto_increment primary key,
    category_name varchar(10) not null unique);
```

--4 Seller (strong entity)

```
Create table Seller_Phone (
    seller_id int not null,
    phone varchar(15) not null,
    Primary key (seller_id, phone),
    Foreign key (seller_id) references seller (seller_id)
);
```

-- Executing first this table

```
Create table Seller (
    seller_id int auto_increment Primary key,
    seller_name varchar(100) not null,
    email varchar(100) not null unique
);
```

--5 Product (strong entity, with specialization)

```
Create table Product (
    product_id int auto_increment primary key,
    product_name varchar(150) not null,
    price decimal(10,2) not null,
    stock int not null default 0,
    category_id ~~int~~ int ~~not~~ null,
    seller_id int not null,
    Foreign key (category_id) references category (category_id)
    on delete set null,
    Foreign key (seller_id) references Seller (seller_id)
    on delete cascade
);
```

-- Specialization : Electronics (disjoint subtype of product)

```
Create table Electronics (
    product_id int primary key
    warranty_period int,
    brand varchar(50),
    foreign key (product_id) references Product (product_id)
    on delete cascade
);
```

--Specialization : Clothing (disjoint subtype of Product)

```
Create table Clothing (
    product_id int primary key,
    size varchar(10),
    fabric varchar(50),
    Foreign key (product_id) references Product (product_id)
    on delete cascade
);
```

--6 Order (strong entity)-

```
Create table Orders(
    order_id int auto-increment primary key,
    order_date Date not null,
    total_amount Decimal(10,2) Not Null,
    customer_id int not null,
    Foreign key (Customer_id) References Customer (customer_id)
    on delete cascade
);
```

--7 Order_item (weak entity depends upon order)

```
Create table Order_item (
    order_id int not null,
    item_no int not null,
    quantity int not null,
    unit_price Decimal(10,2) not null,
    product_id not null,
    primary key (order_id , item_no)
    Foreign key (order_id) references orders (order_id)
    on delete Cascade,
    Foreign key (product_id) references Product
    (product_id) on delete
    set null) ;
```

```
--8 Payment (strong entity, 1:1 with orders)

(create table Payment(
    payment_id int auto_increment primary key,
    amount Decimal(10,2) not null,
    payment_date Date Not null,
    payment_mode varchar(50) not null,
    order_id int not null unique)
    Foreign key(order_id) references Orders(order_id)
    on delete cascade
    );

--9 Delivery (strong entity, 1:1 with orders)
    Create table Delivery (
    delivery_id int auto_increment primary key,
    delivery_date Date,
    status varchar(20) not null,
    tracking_number varchar(30) unique,
    order_id int not null unique,
    Foreign key(order_id) references (order_id)
    on delete cascade
)

-- Sample Data
    Insert into Customer (first_name, Last_name, email) Values
    ('Anuj', 'Saini', 'anujsaini@example.com'),
    ('Yash', 'Singh', 'yashsingh@examples.com');
```

```
Insert into Customer_Phone (Customer_id, phone) Values
    (1, '1259684291'),
    (2, '2596842910');
    (
```

```
Insert into Address (customer_id, address_id, street, city, state, pincode) Values
 (1,1,'12a','Lucknow','Uttar Pradesh','226017'),
 (2, 1, '13b', 'Kanpur', 'Uttar Pradesh, '226172');
```

```
Insert into Category (Category_name) Values
 ('Electronics', 'Clothing', 'Bills');
```

```
Insert into seller (seller_name, email) Values
  ('Cmart Pvt Ltd', 'contact14@gmail.com'),
  ('Fashion Pvt Ltd', 'support@gmail.com');
```

```
Insert into seller_Phone (Seller_id, phone) Values
  (1, '0011-2468'),
  (2, '1100-81842');
```

```
Insert into Product (product_name, price, ~~state~~ stock, category_id) Values
  ('wireless ~~expt~~ cable', '240', 241, 50),
  ('wearable watch', '5000', 4590, 4);
```

```
Insert into Payment (amount, payment_date, payment_mode, order_id)
 Values (3398.00, '2026-08-01', 'UPI', 2);
```

--Demonstrating Referential Integrity

--a) Insert violation : order_id 999 does not exist in Orders

--Expected : Error 1452 - cannot add or update a child row : a foreign key constraint ~~key~~ fails (order_id.FK)

--Insert into Order_item (order_id, item_no, quantity, unit ~~phone~~ price, product_id)

--Values (999, 1, 2, 500.00, 1);

--b Insert Violation: product_id 999 does not exist in Product

--Expected error 1452 - foreign key constraint fails (product_id, FK)

--Insert into Order_item (order_id, item_no, quantity, unit price, product_id)

--Values (1, 3, 1, 100.00, 992);

-- c Unique Violation: this order already has a payment (1:1 constraint)

--Expected: Error 1062 - duplicate entry '1' for key 'order_id'

--Insert into Payment (amount, payment date, payment-mode, order_id)

--Values (24.99, '2026-01-02', 'Online', 2);

--d On delete Set Null demonstration : delete a category

--expected : Product.category_id for its product becomes Null,

--delete from category where category_id = 1;

--Select * from Product where product_id = 1; --category is now null.

--e Not null violation: order date is required

--expected: error 1364 - Field 'order_date' does not have a default value.

--insert into orders (total_amount, customer_id) values (500.00, 1);

19/01/26